

src/components/CodeCityMinimapPortal.tsx

Kind: added · Baseline: f7c2bfd81af2 · Target: NovaDiff · Generated: 2026-05-21T11:00:11.580Z

Overview

`src/components/CodeCityMinimapPortal.tsx` introduces a React portal that mounts a custom minimap into the react-flow controls host. The component hides the built-in minimap, attaches itself to the host, and emits a `code-city-dock-host-ready` event when the attachment succeeds. The implementation relies on DOM queries, a `MutationObserver`, and a retry interval to locate the host element.

Key changes

- **Imports** (R1-R3): `useLayoutEffect`, `useRef`, `useState`, `ReactNode` from `react`; `createPortal` from `react-dom`; `CodeCityMinimapSlot` from the local slot component.
- **Constants** (R5-R6): `COVER_ROOT = ".knowledge-graph-cover-stage"` and `DOCK_HOST_CLASS = "code-city-graph-dock-host"`.
- **Helper functions** (R9-R22):
 - `findControlsHost()` locates the controls panel in the cover stage.
 - `hideBuiltInMinimap()` sets the built-in minimap's display to `none`.
- **Exported component** (R27-R102):
 - Uses `useState` to track the host element.
 - `tryAttach()` performs host lookup, hides the built-in minimap, adds a CSS class, and dispatches the custom event.
 - `useLayoutEffect` sets up the attachment logic, a retry interval, and a `MutationObserver` to react to DOM changes.
 - Cleanup removes the class, restores the minimap display, and disconnects observers.
 - Renders a `<CodeCityMinimapSlot>` inside a portal when the host is available; otherwise falls back to an anchored `<div>`.

Impact

The portal replaces the default minimap with a custom one, ensuring it appears in the bottom-left controls area. It also signals readiness via a custom event, allowing other parts of the application to react once the minimap is docked.

Risks & follow-ups

- **Host element missing:** If the selector `${COVER_ROOT} .react-flow__controls.react-flow__panel.bottom.left` does not match, the component will never attach.
- **Event name collision:** The custom event `code-city-dock-host-ready` could clash with other listeners.
- **Cleanup reliability:** The interval and observer must be cleared on unmount; any failure could leave stale listeners.

- **Styling side-effects:** Hiding the built-in minimap via `display: none !important` may interfere with other CSS rules.

Follow-up actions include adding unit tests for host detection, verifying event dispatch, and ensuring the cleanup logic runs under all unmount scenarios.